Session 25: Equity-Settled Share-Based Payments I: IFRS 2 Mechanics

By the end of this session you must know the scope of IFRS 2 and how equity-settled differs from cash-settled share-based payment, how to measure and recognise the cost of stock options over the vesting period, the distinction between vesting and non-vesting conditions (and between market and non-market conditions), and how to handle forfeiture and expiry of options.

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1. IFRS 2 Scope and the Two Types of Share-Based Payment

- Share-based payments are regulated by IFRS 2, which applies only to share-based payments made for the acquisition of goods or services. It does **not** apply to dividends or to a general issue of shares (e.g. a rights issue for cash).

Share-based payment transaction: a transaction in which an entity receives goods or services in exchange for either (a) equity instruments, or (b) incurring a liability.

Equity-settled share-based payment: the entity issues its own shares (or share options) as payment for goods or services. The goods/services received are measured at fair value and recorded as an asset or expense, with a corresponding increase in equity.

Cash-settled share-based payment: the entity agrees to transfer cash or other assets, in amounts based on the price of its own shares, in return for goods/services received.

2. Measurement: Goods/Services vs. Equity Instruments

Measurement hierarchy: transactions where goods/services are received for equity instruments are, in principle, measured at the fair value of the goods or services received. Only if that fair value cannot be measured reliably is the fair value of the equity instruments granted used instead.

- Timing of the fair value estimate also depends on which measurement basis applies: for transactions measured at the fair value of the goods/services received, fair value is estimated at the **date of receipt**. For transactions measured at the fair value of the equity instruments granted, fair value is estimated at **grant date**.

Equity-settled example: on 1 December 2010, Company A issues 10,000 shares to Company B for goods with a fair value of €15,000. The goods sit in inventory at year-end and are sold for €20,000 during 2011. Entries: at 1 December 2010 (receipt), Dr Inventory 15,000, Cr Share capital 15,000. During 2011 (sale): Dr COGS 15,000, Cr Inventory 15,000. Dr Cash/Receivables 20,000, Cr Revenue 20,000.

Cash-settled example: same facts, but Company A instead pays cash based on the fair value of 10,000 of its own equity shares, which was €15,000 at 1 December 2010 (receipt date) but had risen to €17,000 by 31 December 2010 (year-end). Entries: at receipt, Dr Inventory 15,000, Cr Trade payables 15,000. At year-end, the liability is remeasured to its current fair value: Dr Loss 2,000, Cr Trade payables 2,000 (payable now 17,000). During 2011 (payment and sale): Dr Trade payables 17,000, Cr Cash 17,000. Dr COGS 15,000, Cr Inventory 15,000. Dr Cash/Receivables 20,000, Cr Revenue 20,000.

The key structural difference between the two examples is that a cash-settled liability is **remeasured to fair value at each**

reporting date until settlement (generating a further P&L gain/loss each period), whereas an equity-settled transaction's initial measurement, once made, does **not** get remeasured for subsequent share price movements. This distinction becomes central again in Section 5.

3. The Anatomy of an Executive Stock Option Scheme

- Executive stock option schemes are the most common form of equity-settled share-based payment, giving executives the right to buy shares at a predetermined (exercise) price over a specified period. Often used by growth companies (e.g. technology) to motivate employees, since the option becomes valuable if the share price climbs above the exercise price during the option's life.

Three key dates: the **grant date** (when the option is awarded). The **vesting (service) date** (the end of the service period over which employee services are consumed, e.g. 3 years), and the **maturity date** (the end of the period during which the vested option can be exercised, assuming any performance conditions were satisfied).

4. Vesting Conditions vs. Non-Vesting Conditions

Vesting condition: a condition that determines whether the entity receives the services entitling the counterparty to the share-based payment.

Must know (the vesting-condition taxonomy):
• A vesting condition is either a **service condition** (completing a specified period of service, explicit or implicit) and/or a **performance condition** (meeting specified performance targets while rendering service)
• A performance target is a **non-market condition** if defined by reference to the entity's own (or a group entity's) operations/activities
• A performance target is a **market condition** if defined by reference to the price/value of the entity's (or a group entity's) equity instruments, including shares and share options

Decision test (is a condition a vesting condition at all?): does the condition determine whether the entity receives the services entitling the counterparty to the payment? If no, it is a **non-vesting condition**. If yes, and it requires only a specified period of service, it is a **service condition**. If it requires more (a performance target), classify it as a **non-market** or **market** performance condition per the taxonomy above.

5. Recognition and Measurement of Stock Option Cost

Must know (IFRS 2's recognition timing rule):
• **No vesting conditions:** the employee expense is recognised **immediately**, at grant date
• **Vesting conditions exist:** the expense is recognised **over the vesting period** (e.g. the minimum required service period)

Must know (IFRS 2's measurement rule):
• The cost is measured as the **fair value of the share options at grant date**: fixed once, at the start
• **Market conditions** occurring after the grant date (e.g. subsequent share price movements) are **not** taken into account when determining the P&L charge over the vesting period. The grant-date fair value estimate already builds in the probability distribution of future share prices, so it is not revisited

6. Worked Examples: No Vesting Conditions vs. a Service Condition

No vesting conditions (Example 1, ordinary shares): on 31 December 20 X 1, Entity A grants each employee 10 ordinary shares, with no vesting conditions. Because employees have an unconditional right to the shares immediately at grant date, Entity A recognises the full staff cost (and the corresponding increase in equity) on 31 December 20 X 1 itself.

No vesting conditions (Example 2, options): Entity A grants 10 share options per employee to reward past performance, exercisable any time after 31 December 20 X 2, with no further conditions to be met (even an employee who leaves before that date keeps the right to exercise later). Because there is no vesting condition (departing employees are not excluded), the full cost is again recognised at grant date, 31 December 20 X 1.

With a service condition (Example 3): Entity A grants 10 share options per employee, but exercise is conditional on the employee working throughout 20 X 2. A service condition. Because the services being paid for are rendered over 20 X 2 (the vesting period), Entity A recognises the staff cost and the corresponding equity increase during 20 X 2 (not at the 31 December 20 X 1 grant date).

Measuring at grant-date fair value (Example 4): Entity A grants employees ordinary shares worth €5,000 in total fair value at grant date (31 December 20 X 1), with no vesting conditions. The full expense is recognised immediately: Dr Profit or loss (staff expense) 5,000. Cr Equity 5,000. The measurement basis is explicitly the fair value of the shares (not of the employee services), at grant date.

7. Full Worked Example: A Multi-Year Vesting Schedule

Setup: on 1 January 2011, M plc grants each of 4 executives options to purchase 25,000 shares at 25 cents/share (the exercise price). Share value at grant date = €1. Nominal value of shares = 10 cents. Fair value of each option at grant date = 75 cents. The vesting condition requires 3 years of service from 1 January 2011.

Annual expense (Solution 5): total grant-date fair value of all options = 4 executives × 25,000 options × €0.75 = €75,000, spread evenly over the 3-year vesting period: €75,000 ÷ 3 = **€25,000 per year**. Journal entry each of 31 December 2011, 2012 and 2013: Dr Share option expense 25,000. Cr Equity (stock options) 25,000. Note that the subsequent share price rises (€1.50 in 2012, €2 in 2013, €3 in 2014) are **irrelevant** to this calculation, per Section 5's rule that market conditions after grant date do not affect the P&L charge.

Exercise (Solution 5, continued): if all four executives exercise in full on 1 January 2014, each pays 25,000 × €0.25 = €6,250 (total cash received = 4 × 6,250 = €25,000). Entries at exercise: Dr Cash 25,000. Dr Equity (stock options) 75,000 (the full accumulated balance built up over the 3 years). Cr Ordinary share capital 10,000 (= 4 × 25,000 × €0.10 nominal value). Cr Share premium 90,000 (the balancing figure).

8. Forfeiture and Expiry

Must know (what happens when a vesting condition is not met):
• **Failure to meet a service condition or a non-market performance condition:** the expense is **revised** to reflect the best available estimate of the number of equity instruments actually expected to vest (i.e. forfeited awards are backed out of the cumulative expense)
• **Failure to meet a market condition: no change** to the accounting. The entity continues recognising the expense over the remainder of the vesting period exactly as originally planned, since market-condition probabilities were already built into the grant-date fair value

Example 6, one executive leaves during the vesting period (Solution 6): using the same facts as Example 5, but one of the four executives leaves on 1 January 2012, losing their options (a service-condition forfeiture, so the estimate is revised). 31 December 2011 charge remains €25,000 (before the departure is known, since it occurs on 1 January 2012). 31 December 2012 charge = 12,500. Computed either as [(3 executives × 25,000 × 0.75 × 2/3) − 25,000 already recognised] = 12,500, or equivalently per-executive: 25,000÷4 = 6,250/year-executive, so (3 × 6,250) − 6,250 = 12,500. 31 December 2013 charge = 18,750 (= 3 × 6,250, the remaining 3 executives' full annual share). At exercise (for the 3 remaining executives): Dr Cash 18,750 (= 3 × 25,000 × 0.25). Dr Equity (stock options) 56,250 (= 25,000 + 12,500 + 18,750, the cumulative balance). Cr Ordinary share capital 7,500 (= 3 × 25,000 × 0.10). Cr Share premium 67,500 (balancing figure).

What if vested options simply expire unexercised (Solution 6, continued)? after the exercise period passes with no exercise, the accumulated equity balance is simply reclassified within equity: Dr Equity (stock options) 56,250. Cr Equity (expired stock options) 56,250. There is **no change** to the company's assets, liabilities or total equity. The expense already recognised over the vesting period is not reversed merely because the vested option was never exercised.

8.1 New Example 7: Benning and an uncertain vesting date

The revised handout adds a non market performance condition. The fair value per share stays fixed at grant date, but both the expected employee count and expected vesting period can change. Recompute **cumulative** expense at each year end, then subtract expense already booked to find the current year's charge.

Grant facts. Each of 50 employees receives 100 shares with a grant date fair value of EUR 20 each, conditional on staying until vesting. Vesting is after year 1 if earnings growth exceeds 18%, after year 2 if average annual growth exceeds 13%, or after year 3 if average annual growth exceeds 10%. These targets concern company earnings, not its share price, so they are non market conditions.

Year 1. Earnings grow 14%, so the year 1 condition fails. Three employees have left, and three more are expected to leave before expected vesting at the end of year 2. Expected awards are 44 × 100 shares. Total expected cost =44 × 100 × 20 = **EUR 88,000**. One of the expected two service years has been completed: cumulative cost =88,000 × 1/2 = **EUR 44,000**. Annual charge is 44,000.

Year 2. Earnings grow 10%. The source uses the arithmetic average: (14% +10%)/2 = 12%, below 13%, so the shares have not vested. Two more employees leave. Two additional departures are now expected in year 3, giving 43 expected remaining employees. Expected total cost = 43 × 100 × 20 = **EUR 86,000**. Two of three expected service years are complete: cumulative cost = 86,000 × 2/3 = **EUR 57,333.33**. Deduct the 44,000 already booked. Year 2 expense is **EUR 13,333.33**.

Year 3. Earnings grow 8%. Average growth over the three years is (14% +10% +8%)/3 = 10.6667%, above 10%, so the condition is satisfied. Three more employees actually leave. Actual remaining employees = 50 minus 3 minus 2 minus 3 = **42**. All service is complete. Final cumulative cost = 42 × 100 × 20 = **EUR 84,000**. Year 3 expense = 84,000 minus 57,333.33 = **EUR 26,666.67**.

Year	Expected or actual employees	Expected vesting period	Cumulative expense EUR	Annual expense EUR
1	44 expected	2 years	44,000.00	44,000.00
2	43 expected	3 years	57,333.33	13,333.33
3	42 actual	3 years, now vested	84,000.00	26,666.67

Each annual entry debits share based payment expense and credits equity for that year's charge. The annual amounts total 84,000. A change in expected vesting date does not justify leaving the old allocation unchanged, and the year 2 estimate must not use the year 3 actual employee count before it is known. The calculations above are derived from Example 7 facts on slides 39 to 42. The new slides ask for a solution rather than printing one.

9. Disclosure Requirements

Must know (main IFRS 2 disclosures):

- The nature and extent of share-based payment arrangements that existed during the period
- How the fair value of the goods/services received, or of the equity instruments granted, was determined
- The effect of share-based payment transactions on the period's profit or loss and on financial position

10. Common Mistakes and Exam Traps

- Remeasuring an equity-settled option's cost for subsequent share price movements. Grant-date fair value is fixed once and never revisited for market-condition changes. Only cash-settled liabilities are remeasured to current fair value at each reporting date.
- Recognising the full option cost at grant date whenever a service condition exists. A genuine vesting condition (service or non-market performance) spreads the cost over the vesting period. Only the absence of any vesting condition triggers immediate full recognition.
- Reversing previously-recognised expense when a vested option later expires unexercised. Expiry after vesting is a pure equity reclassification with no P&L or asset/liability effect. Only forfeiture *during* the vesting period (failing a service/non-market condition) revises the cumulative expense.
- Revising the cumulative expense for a failed market condition (e.g. a share-price target not met). Market-condition failures get no accounting adjustment at all, unlike service or non-market performance condition failures.
- Confusing which figure is the balancing figure in the exercise-date journal entry. Cash received and share capital (at nominal value) are both directly computable. Share premium is always the plug that makes the entry balance.

11. Must Know Absolutely

Must know:

• IFRS 2 covers share-based payment for goods/services only, not dividends or general share issues

• Equity-settled: measure goods/services fair value (or equity instrument fair value if unreliable). Cash-settled: recognise a liability, remeasured to fair value each period until settlement

• No vesting condition → full expense at grant date. Vesting condition (service and/or performance) → expense spread over the vesting period

• Cost is always measured at the fair value of the equity instrument **at grant date**, fixed regardless of later market-condition/share-price movements

• Forfeiture (service/non-market condition failure) → revise cumulative expense to the revised number expected to vest. Market-condition failure → no change. Expiry after vesting → pure equity reclassification, zero P&L/net-asset effect